

Администрирование локальных сетей

Лабораторная работа №5

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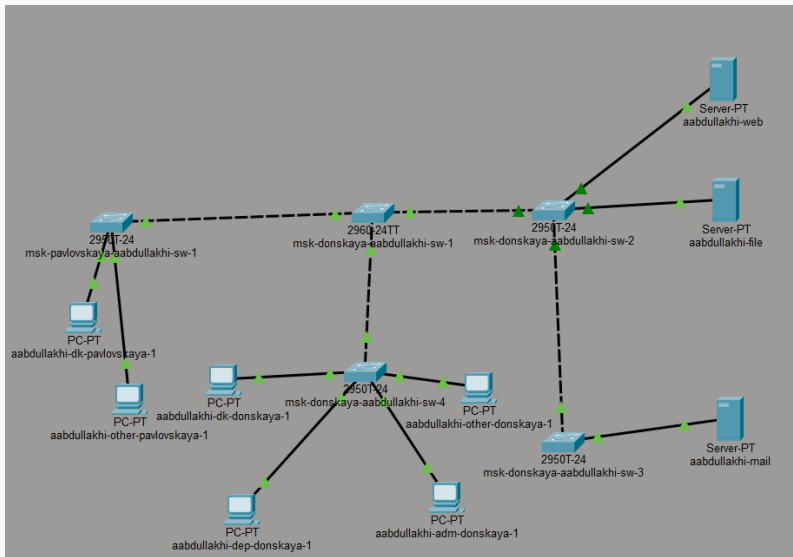
27 февраля 2026

Российский университет дружбы народов, Москва, Россия

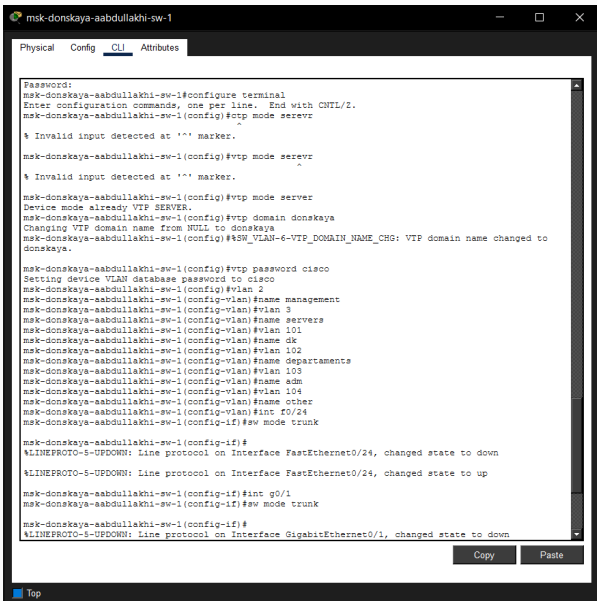
Цель

Основная цель работы — получение практических навыков конфигурации виртуальных локальных сетей (VLAN) на управляемых коммутаторах.

Выполнение лабораторной работы



Настройка VTP-сервера и создание VLAN



```
msk-donskaya-aabdullakhi-sw-1
Physical Config CLI Attributes
Password:
msk-donskaya-aabdullakhi-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aabdullakhi-sw-1(config)#vtp mode server
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-sw-1(config)#vtp mode server
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-sw-1(config)#vtp mode server
Device mode already VTP SERVER.
msk-donskaya-aabdullakhi-sw-1(config)#vtp domain donskeya
Changing VTP domain name from NULL to donskeya
msk-donskaya-aabdullakhi-sw-1(config)#vtp domain donskeya
VTP domain name changed to donskeya.

msk-donskaya-aabdullakhi-sw-1(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-aabdullakhi-sw-1(config)#vlan 2
msk-donskaya-aabdullakhi-sw-1(config-vlan)#name management
msk-donskaya-aabdullakhi-sw-1(config-vlan)#vlan 3
msk-donskaya-aabdullakhi-sw-1(config-vlan)#name servers
msk-donskaya-aabdullakhi-sw-1(config-vlan)#vlan 101
msk-donskaya-aabdullakhi-sw-1(config-vlan)#name dk
msk-donskaya-aabdullakhi-sw-1(config-vlan)#vlan 102
msk-donskaya-aabdullakhi-sw-1(config-vlan)#name departments
msk-donskaya-aabdullakhi-sw-1(config-vlan)#vlan 103
msk-donskaya-aabdullakhi-sw-1(config-vlan)#name adm
msk-donskaya-aabdullakhi-sw-1(config-vlan)#vlan 104
msk-donskaya-aabdullakhi-sw-1(config-vlan)#name other
msk-donskaya-aabdullakhi-sw-1(config-vlan)#int f0/24
msk-donskaya-aabdullakhi-sw-1(config-if)#sw mode trunk

msk-donskaya-aabdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

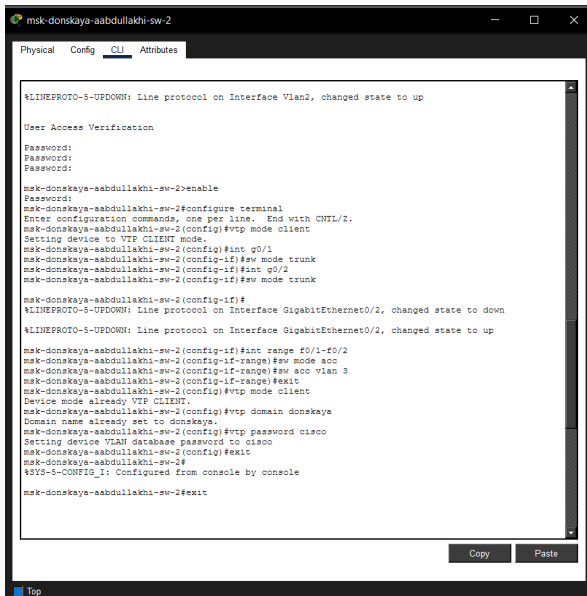
msk-donskaya-aabdullakhi-sw-1(config-if)#int g0/1
msk-donskaya-aabdullakhi-sw-1(config-if)#sw mode trunk

msk-donskaya-aabdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
```

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настройка ведомых коммутаторов
(sw-2, sw-3, sw-4, pavlovskaya-sw-1)



```
msk-donskaya-aabdullakhi-sw-2
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

User Access Verification
Password:
Password:
Password:

msk-donskaya-aabdullakhi-sw-2>enable
Password:
msk-donskaya-aabdullakhi-sw-2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aabdullakhi-sw-2(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-aabdullakhi-sw-2(config)#int g0/1
msk-donskaya-aabdullakhi-sw-2(config-if)#sw mode trunk
msk-donskaya-aabdullakhi-sw-2(config-if)#int g0/2
msk-donskaya-aabdullakhi-sw-2(config-if)#sw mode trunk

msk-donskaya-aabdullakhi-sw-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

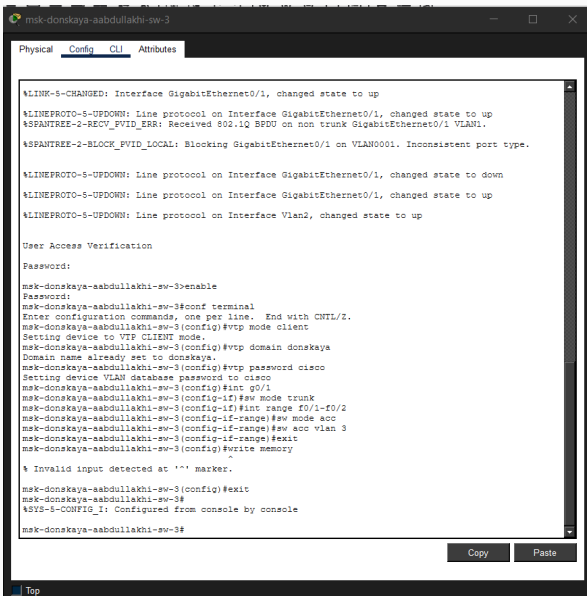
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

msk-donskaya-aabdullakhi-sw-2(config-if)#int range f0/1-f0/2
msk-donskaya-aabdullakhi-sw-2(config-if-range)#sw mode acc
msk-donskaya-aabdullakhi-sw-2(config-if-range)#sw acc vlan 3
msk-donskaya-aabdullakhi-sw-2(config-if-range)#exit
msk-donskaya-aabdullakhi-sw-2(config)#vtp mode client
Device mode already VTP CLIENT.
msk-donskaya-aabdullakhi-sw-2(config)#vtp domain dontkaya
Domain name already set to dontkaya.
msk-donskaya-aabdullakhi-sw-2(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-aabdullakhi-sw-2(config)#exit
msk-donskaya-aabdullakhi-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aabdullakhi-sw-2#exit
```

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```
msk-donskaya-aabdullakhi-sw-3
Physical Config CLI Attributes

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SPANTREE-2-RECV_FVID_ERR: Received 802.1Q BPDU on non trunk GigabitEthernet0/1 VLAN1.
%SPANTREE-2-BLOCK_FVID_LOCAL: Blocking GigabitEthernet0/1 on VLAN0001. Inconsistent port type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

User Access Verification
Password:

msk-donskaya-aabdullakhi-sw-3>enable
Password:
msk-donskaya-aabdullakhi-sw-3#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aabdullakhi-sw-3(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-aabdullakhi-sw-3(config)#vtp domain dontskaya
Domain name already set to dontskaya.
msk-donskaya-aabdullakhi-sw-3(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-aabdullakhi-sw-3(config)#int g0/1
msk-donskaya-aabdullakhi-sw-3(config-if)#sw mode trunk
msk-donskaya-aabdullakhi-sw-3(config-if)#int range f0/1-f0/2
msk-donskaya-aabdullakhi-sw-3(config-if-range)#sw mode acc
msk-donskaya-aabdullakhi-sw-3(config-if-range)#sw acc vlan 3
msk-donskaya-aabdullakhi-sw-3(config-if-range)#exit
msk-donskaya-aabdullakhi-sw-3(config)#write memory

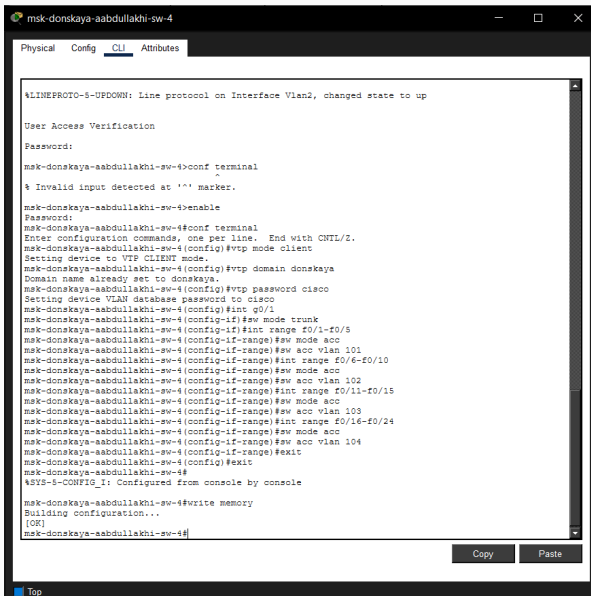
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-sw-3(config)#exit
msk-donskaya-aabdullakhi-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aabdullakhi-sw-3#
```

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```
msk-donskaya-aabdullakhi-sw-4
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

User Access Verification

Password:

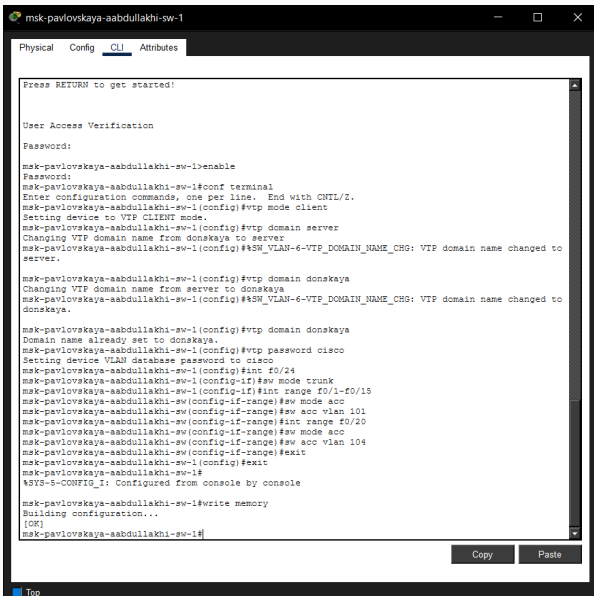
msk-donskaya-aabdullakhi-sw-4>conf terminal
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-sw-4>enable
Password:
msk-donskaya-aabdullakhi-sw-4#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aabdullakhi-sw-4(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-aabdullakhi-sw-4(config)#vtp domain donskaya
Domain name already set to donskaya.
msk-donskaya-aabdullakhi-sw-4(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-aabdullakhi-sw-4(config)#int g0/1
msk-donskaya-aabdullakhi-sw-4(config-if)#sw mode trunk
msk-donskaya-aabdullakhi-sw-4(config-if)#int range f0/1-f0/5
msk-donskaya-aabdullakhi-sw-4(config-if-range)#sw mode acc
msk-donskaya-aabdullakhi-sw-4(config-if-range)#sw acc vlan 101
msk-donskaya-aabdullakhi-sw-4(config-if-range)#int range f0/6-f0/10
msk-donskaya-aabdullakhi-sw-4(config-if-range)#sw mode acc
msk-donskaya-aabdullakhi-sw-4(config-if-range)#sw acc vlan 102
msk-donskaya-aabdullakhi-sw-4(config-if-range)#int range f0/11-f0/15
msk-donskaya-aabdullakhi-sw-4(config-if-range)#sw mode acc
msk-donskaya-aabdullakhi-sw-4(config-if-range)#sw acc vlan 103
msk-donskaya-aabdullakhi-sw-4(config-if-range)#int range f0/16-f0/24
msk-donskaya-aabdullakhi-sw-4(config-if-range)#sw mode acc
msk-donskaya-aabdullakhi-sw-4(config-if-range)#sw acc vlan 104
msk-donskaya-aabdullakhi-sw-4(config-if-range)#exit
msk-donskaya-aabdullakhi-sw-4(config)#exit
msk-donskaya-aabdullakhi-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aabdullakhi-sw-4#write memory
Building configuration...
[OK]
msk-donskaya-aabdullakhi-sw-4#
```

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```
msk-pavlovskaya-aabdullakhi-sw-1
Physical Config CLI Attributes

Press RETURN to get started!

User Access Verification

Password:

msk-pavlovskaya-aabdullakhi-sw-1>enable
Password:
msk-pavlovskaya-aabdullakhi-sw-1#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-pavlovskaya-aabdullakhi-sw-1(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-pavlovskaya-aabdullakhi-sw-1(config)#vtp domain server
Changing VTP domain name from dontskaya to server
msk-pavlovskaya-aabdullakhi-sw-1(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
server.

msk-pavlovskaya-aabdullakhi-sw-1(config)#vtp domain dontskaya
Changing VTP domain name from server to dontskaya
msk-pavlovskaya-aabdullakhi-sw-1(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
dontskaya.

msk-pavlovskaya-aabdullakhi-sw-1(config)#vtp domain dontskaya
Domain name already set to dontskaya.
msk-pavlovskaya-aabdullakhi-sw-1(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-pavlovskaya-aabdullakhi-sw-1(config)#int f0/24
msk-pavlovskaya-aabdullakhi-sw-1(config-if)#sw mode trunk
msk-pavlovskaya-aabdullakhi-sw-1(config-if)#int range f0/1-f0/15
msk-pavlovskaya-aabdullakhi-sw-1(config-if-range)#sw mode acc
msk-pavlovskaya-aabdullakhi-sw-1(config-if-range)#sw acc vlan 101
msk-pavlovskaya-aabdullakhi-sw-1(config-if-range)#int range f0/20
msk-pavlovskaya-aabdullakhi-sw-1(config-if-range)#sw mode acc
msk-pavlovskaya-aabdullakhi-sw-1(config-if-range)#sw acc vlan 104
msk-pavlovskaya-aabdullakhi-sw-1(config-if-range)#exit
msk-pavlovskaya-aabdullakhi-sw-1(config)#exit
msk-pavlovskaya-aabdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-pavlovskaya-aabdullakhi-sw-1#write memory
Building configuration...
[OK]
msk-pavlovskaya-aabdullakhi-sw-1#
```

Назначение IP-адресов серверам

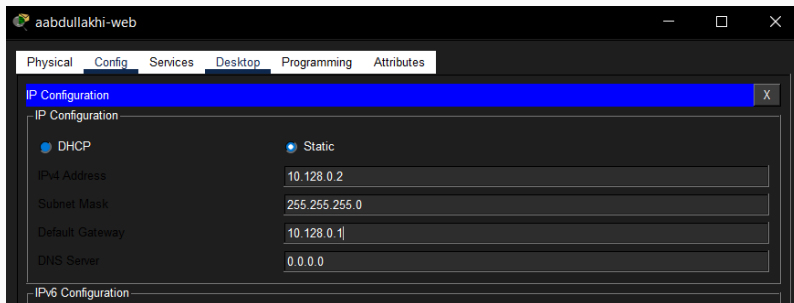


Рис. 7: Web-сервер

Назначение IP-адресов рабочим станциям

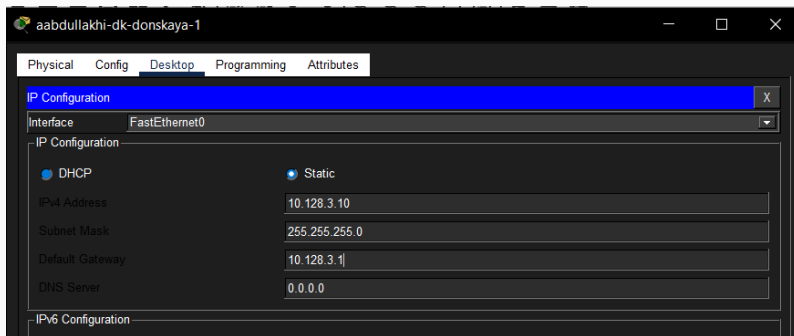
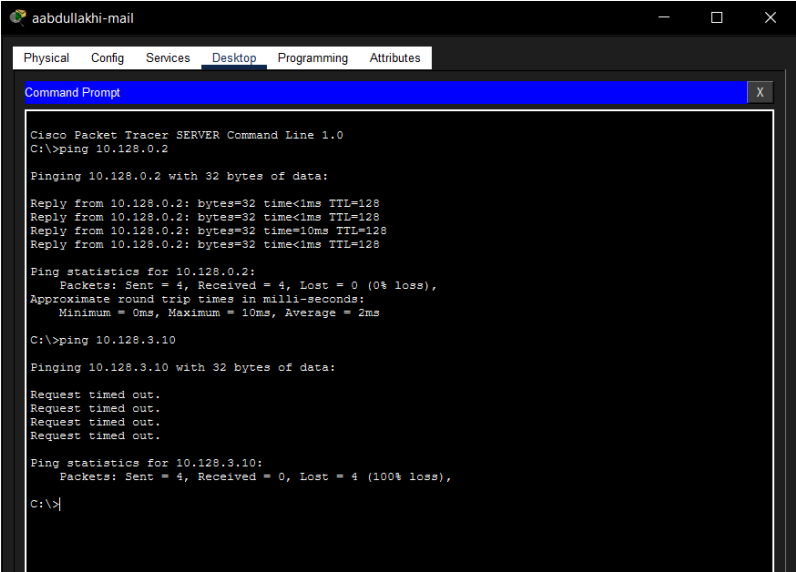


Рис. 8: aabdullakhi-dk-donskaya-1

Верификация сети (Диагностика Ping)



The screenshot shows a Cisco Packet Tracer interface with a 'Command Prompt' window open. The window title is 'aabdullakhi-mail'. The 'Desktop' tab is selected. The Command Prompt displays the output of two ping commands. The first command, 'ping 10.128.0.2', shows successful results with 4 packets received and 0% loss. The second command, 'ping 10.128.3.10', shows failed results with 0 packets received and 100% loss.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Reply from 10.128.0.2: bytes=32 time<1ms TTL=128
Reply from 10.128.0.2: bytes=32 time<1ms TTL=128
Reply from 10.128.0.2: bytes=32 time=10ms TTL=128
Reply from 10.128.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 10.128.3.10

Pinging 10.128.3.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.3.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Анализ трафика в режиме Simulation

Cisco Packet Tracer - C:\Users\WAHID JAN\Cisco Packet Tracer 9.0.0\saves\lab4.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x 524, y 242

Root 14:15:00

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	aabdullakhi-dk-donskaya-1
	0.002	msk-donskaya-aabdullakhi-sw
	0.003	msk-donskaya-aabdullakhi-sw
	0.004	msk-pavlovskaya-aabdullakhi-sw
	0.005	aabdullakhi-dk-pavlovskaya-1
	0.006	msk-pavlovskaya-aabdullakhi-sw
	0.007	msk-donskaya-aabdullakhi-sw
Visible	0.008	msk-donskaya-aabdullakhi-sw
	0.419	--
	0.420	msk-donskaya-aabdullakhi-sw
	0.420	msk-donskaya-aabdullakhi-sw

Reset Simulation ☒ Constant Delay Captured to: 0.420 s

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIP, RIPv2, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Time: 01:12:31.575 PLAY CONTROLS

Event List Realtime Simulation

Детальный разбор структуры пакета

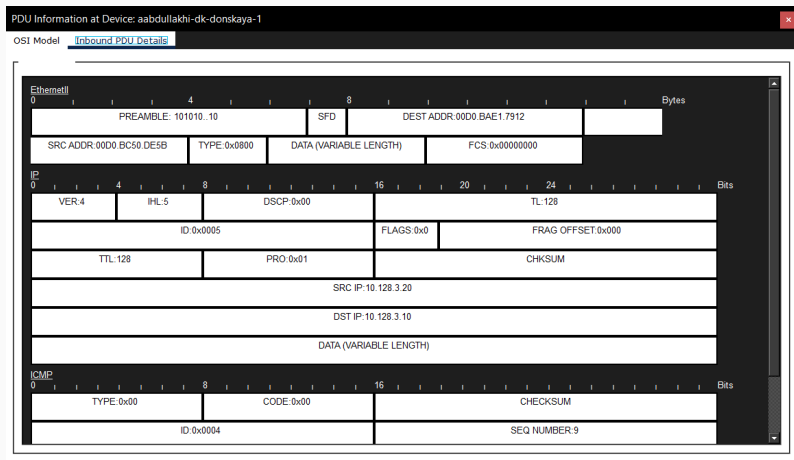


Рис. 11: Детальный разбор структуры пакета

Выводы

В ходе лабораторного практикума была полностью смоделирована и настроена локальная сеть предприятия. На центральном коммутаторе успешно развернут VTP-сервер, что позволило централизованно создать необходимые VLAN и автоматически распространить информацию о них на все клиентские коммутаторы. Проведена сегментация сети на логические домены (управление, серверы, отделы), на конечных узлах настроены IP-адреса.

Вывод
